Lecture 9: Rocky Planets I
Earth & Venus

Planetary Systems

Tim Lichtenberg

Kapteyn Astronomical Institute
University of Groningen

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Three Venus-bound missions over an Akatsuki UV view of Venus. Credit: Widemann et al. (2023), SSRv 219, 56

Learning Objectives

By the end of this lecture, you will be able to:

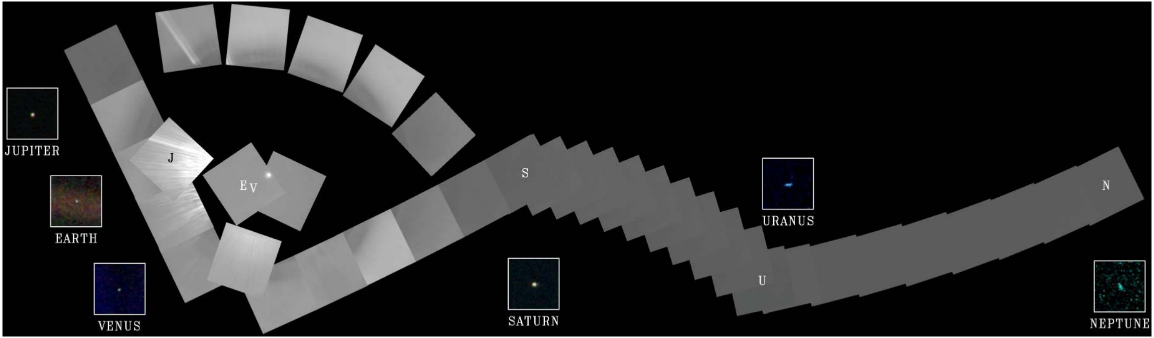
- Use Earth's interior, surface, ocean and climate as the reference case and compare Venus to it
- Derive the Simpson-Nakajima runaway greenhouse limit and explain why a saturated atmosphere has a maximum thermal flux
- Read the D/H ratio as evidence for past water on Venus

Mercury, Venus, Earth and Mars at relative scale: similar accretion zones, radically different surface environments. Credit: NASA/JPL.

Recap: Lectures 1–8

- Differentiation separates crust, mantle, and core ($C/MR^2 = 0.331$ for Earth)
- Volcanism, impact cratering, and escape shape surfaces and atmospheres
- Liquid outer core sustains a protective magnetic dynamo

Today: What caused the evolutionary divergence of Earth and Venus?



Voyager 1 Solar System family portrait (1990), with inset frames for the planets it detected.
Credit: NASA/JPL-Caltech.

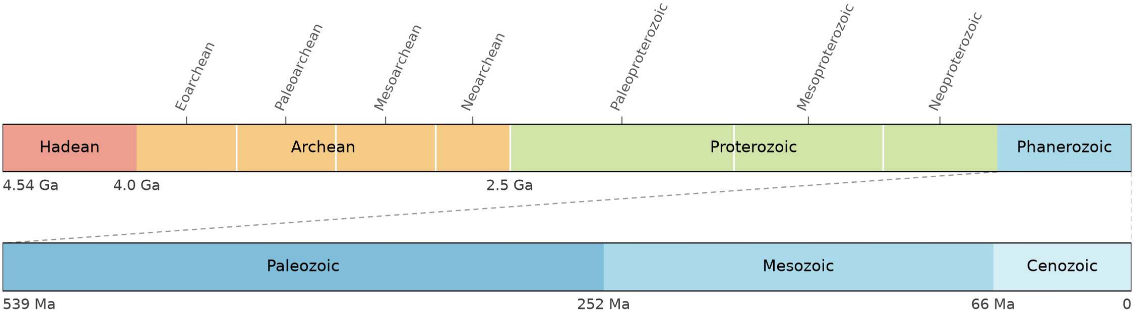
1. Earth as Reference

Earth seen from ~ 6 billion km. Voyager 1 (1990). Credit: NASA/JPL-Caltech

Earth at a Glance

Parameter	Value	Note
Mass	5.972×10^{24} kg	defines $M_\oplus$
Radius (mean)	6371 km	defines $R_\oplus$
Mean density	$5515\ \mathrm{kg\,m^{-3}}$	uncompressed ~ $4030\ \mathrm{kg\,m^{-3}}$
Semimajor axis	1.000 AU	defines AU
Bond albedo	0.30	cloud + ice + land
T_{eq}	255 K	equilibrium, no greenhouse
T_{surf}	288 K	+33 K greenhouse warming
Rotation period	23.93 h	spin axis tilt 23.4°
Magnetic field	dipolar, ~ 25–65 μT	active dynamo

Source: NASA Earth Fact Sheet (NSSDC)



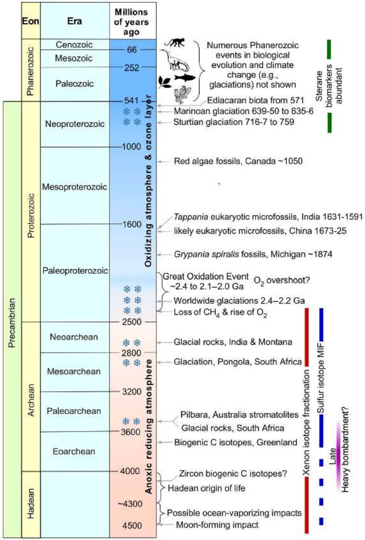
Geologic eons and eras across Earth's 4.54 Gyr; the Precambrian takes almost 90% of the timeline. Credit: Gradstein et al. (2020).

What Makes Earth Habitable?

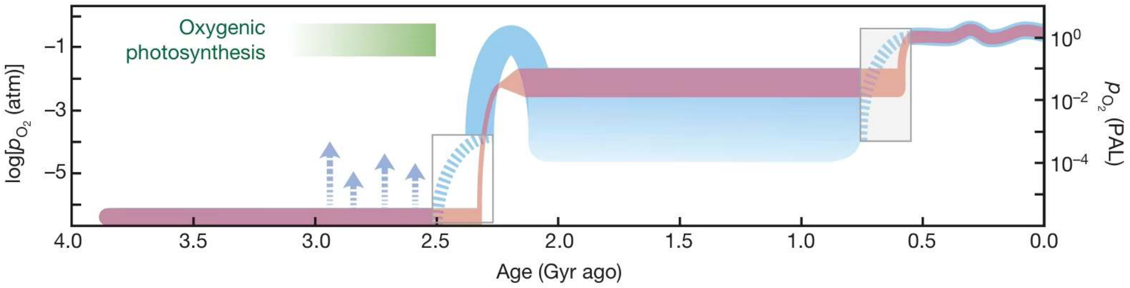
Four ingredients, all present and stable for ~ 4 Gyr:

- **Liquid water** on the surface, continuously
- **Plate tectonics**: long-term carbon cycle, climate buffering
- **Magnetic field** against solar wind erosion, and a **stable orbit** with a large stabilising Moon

Identify which planetary features are necessary, which are sufficient, and which are accidental.

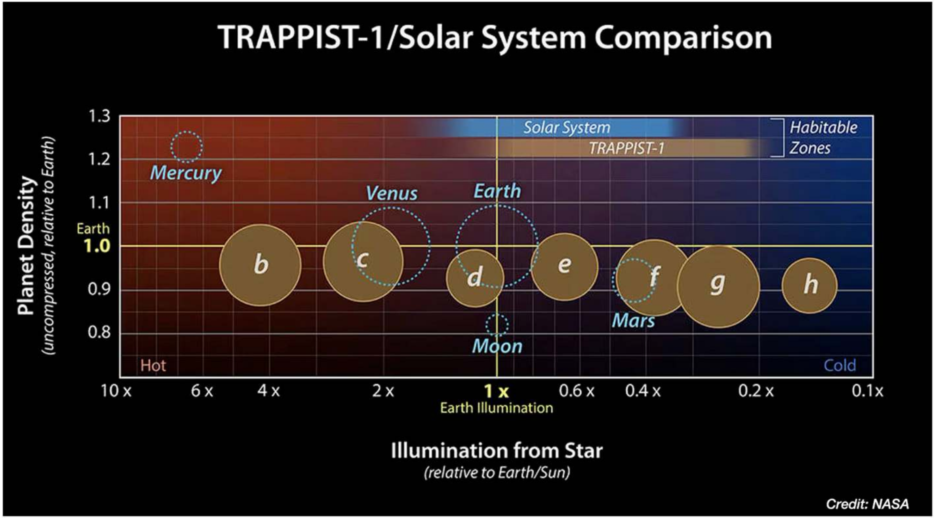


Precambrian milestones from the first oceans through the Great Oxidation Event to the Neoproterozoic glaciations and the ozone layer. Credit: Catling & Zahnle (2020).



Atmospheric oxygen through Earth history, on a logarithmic scale relative to the present level. Reproduced from Lyons et al. (2014).

Anoxic below 10^{-5} of present for about 2 Gyr, an abrupt rise at the Great Oxidation Event near 2.4 Ga, a mid-Proterozoic plateau, and a second rise to modern levels in the Neoproterozoic and Phanerozoic.

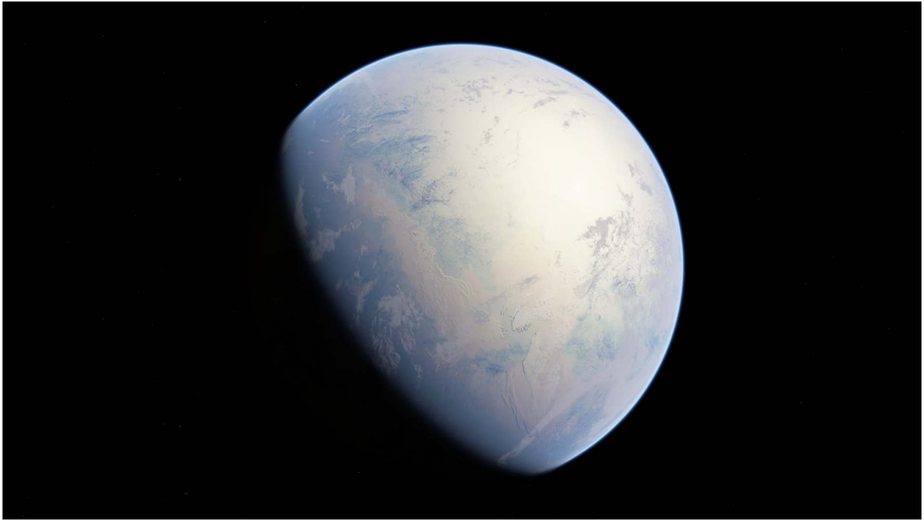


The seven TRAPPIST-1 planets against the inner solar system in density and stellar illumination: three (e, f, g) lie in the habitable zone of a late M dwarf. Credit: NASA/JPL-Caltech (PIA22093).

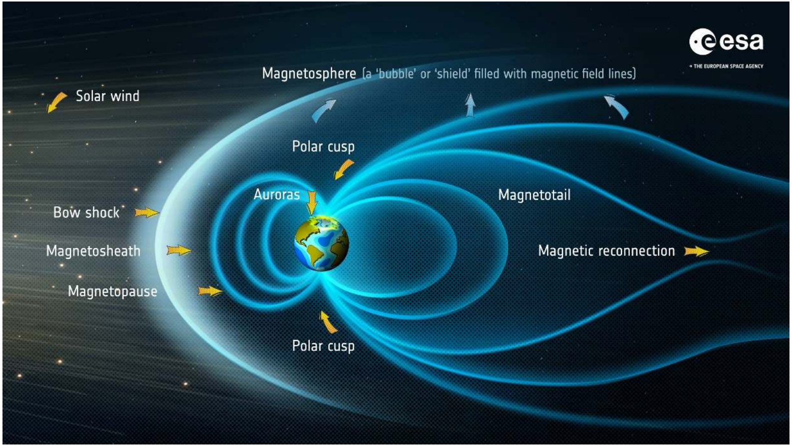
Plate Tectonics: The Engine

Earth alone among known solar system bodies runs plate tectonics today.

- About 15 major plates move at cm yr^{-1} speeds
- Subduction recycles crust and surface volatiles on ~ 100 Myr timescales
- Buried carbonate returns to the mantle: the long-term carbon cycle closes



Cryogenian Snowball Earth: high ice albedo held the freeze until volcanic outgassing and the carbonate-silicate feedback forced deglaciation. Credit: Oleg Kuznetsov, Wikimedia Commons, CC BY-SA 4.0.

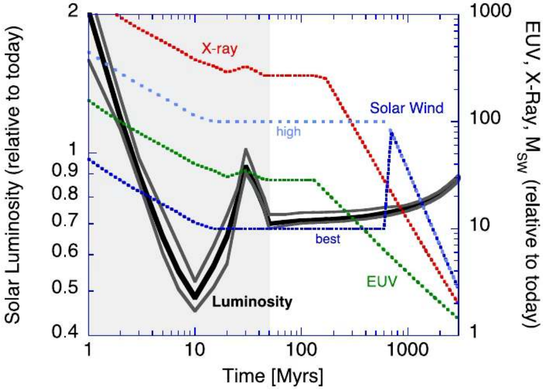


Anatomy of Earth's magnetosphere, from the bow shock to the nightside tail. Credit: ESA, CC BY-SA 3.0 IGO.

A surface dipole of 25–65 $\mu\mathrm{T}$ from core convection holds the dayside magnetopause at ~ 10 $R_{\oplus}$ and suppresses ion-pickup escape of volatiles.

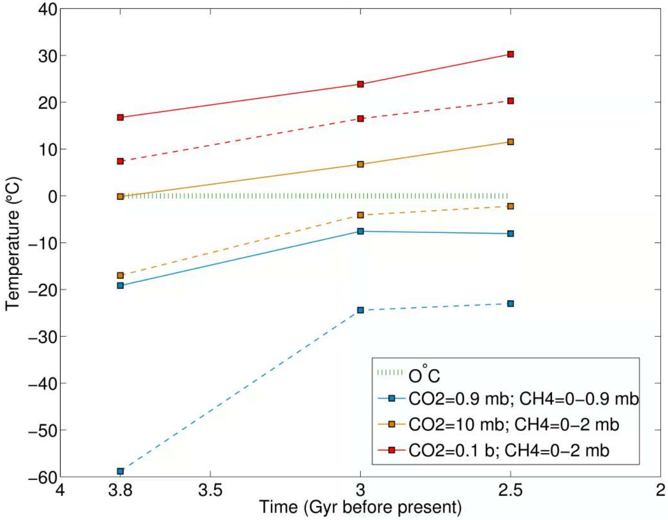
The Blue Marble: Earth's surface ocean and polar cryosphere from Apollo 17. Credit: NASA / Apollo 17 crew.

The oceans hold 1.34 × $10^9\ \mathrm{km}^3$ of water, about 2700 m global equivalent layer (GEL); the deep mantle holds ≥ 1–2 additional ocean masses.



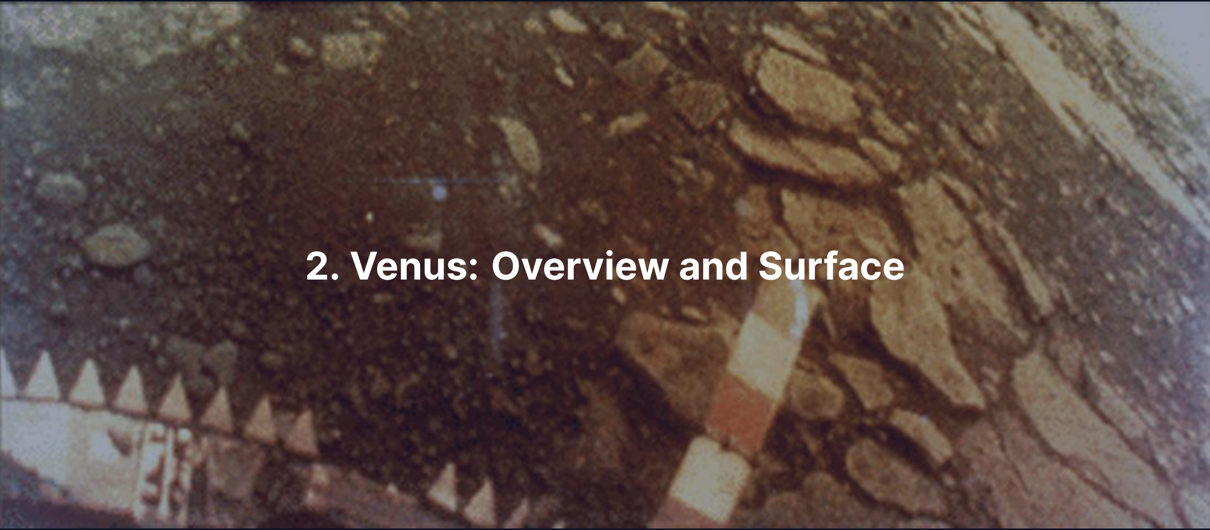
Solar luminosity and the faint young Sun problem over Earth history. Credit: Zahnle et al. (2007).

The Sun brightened by ~ 30% since 4.5 Ga, yet liquid water persisted at ≥ 3.8 Ga: the carbonate-silicate feedback stabilised the climate while the inner edge of the habitable zone swept outward past Venus.

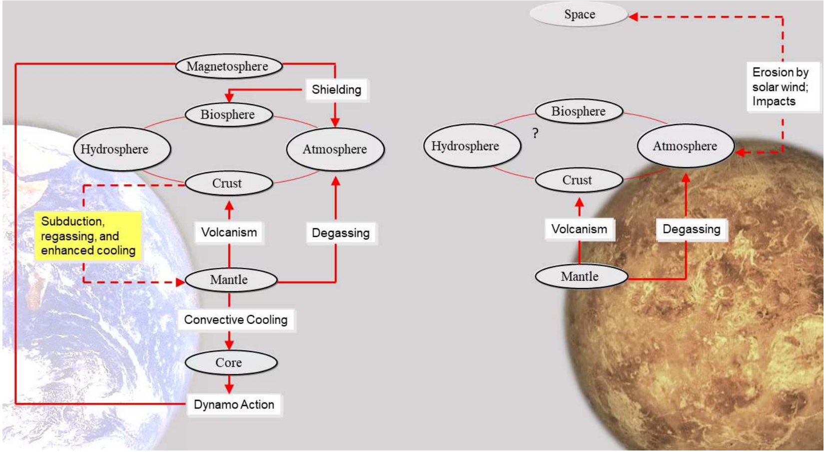


Archean global mean temperature from 3.8 to 2.5 Ga in a 3D model for three CO_2 and CH_4 compositions; about 0.1 bar of CO_2 resolves the faint-young-Sun problem at all three epochs. Reproduced from Charnay et al. (2013).

2. Venus: Overview and Surface



The surface of Venus under its 92-bar atmosphere, Venera 13 (1982). Credit: USSR/NASA NSSDC, public domain



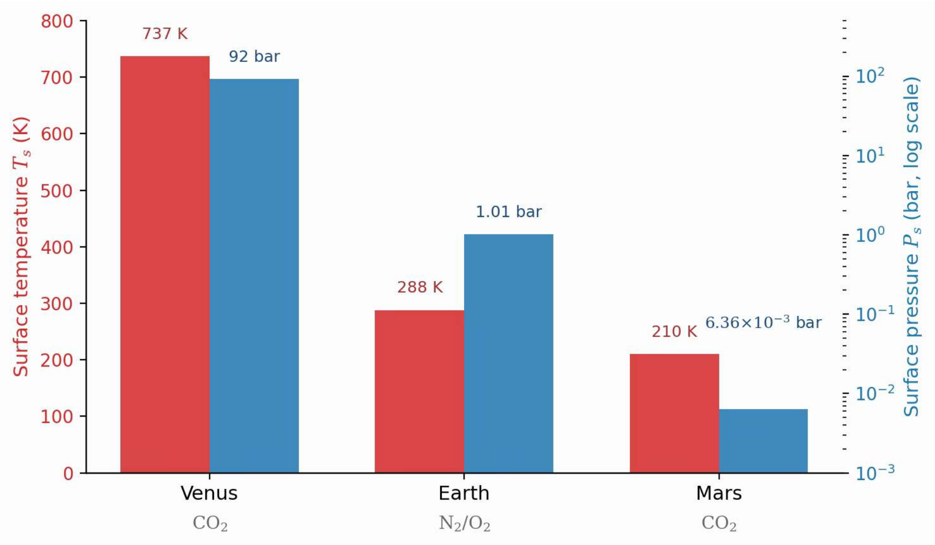
Earth and Venus as coupled systems of interior, magnetosphere, atmosphere and surface. Credit: Gillmann et al. (2022).

The differences between the twins are emergent properties of which feedback loops close.

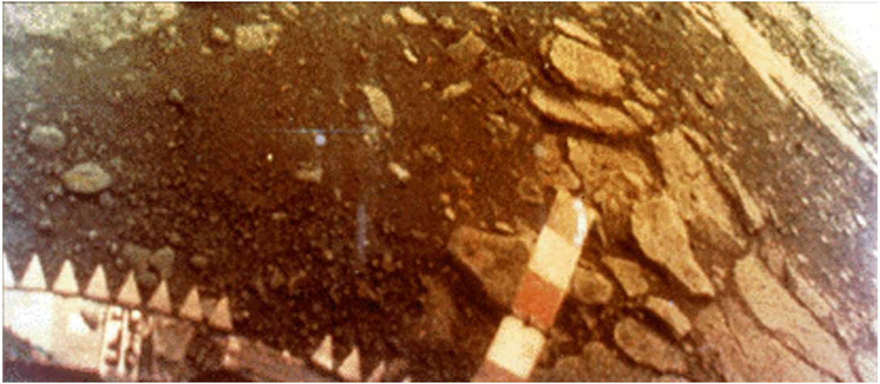
Venus and Earth: Twin and Not

Parameter	Earth	Venus
Mass ($M_\oplus$)	1.000	0.815
Radius ($R_\oplus$)	1.000	0.949
Mean density (kg m^{-3})	5515	5243
Semimajor axis (AU)	1.000	0.723
Stellar flux ($S_\oplus$)	1.00	1.91
Bond albedo	0.30	0.77
$T_{\rm surf}$ (K)	288	737
Surface pressure (bar)	1.0	92
Atmosphere	N_2/O_2	CO_2/N_2
Rotation period (d)	+1.0	-243 (retrograde)
Magnetic field	~ 50 μT	< 10^{-5}× Earth

Source: NASA Venus Fact Sheet (NSSDC)

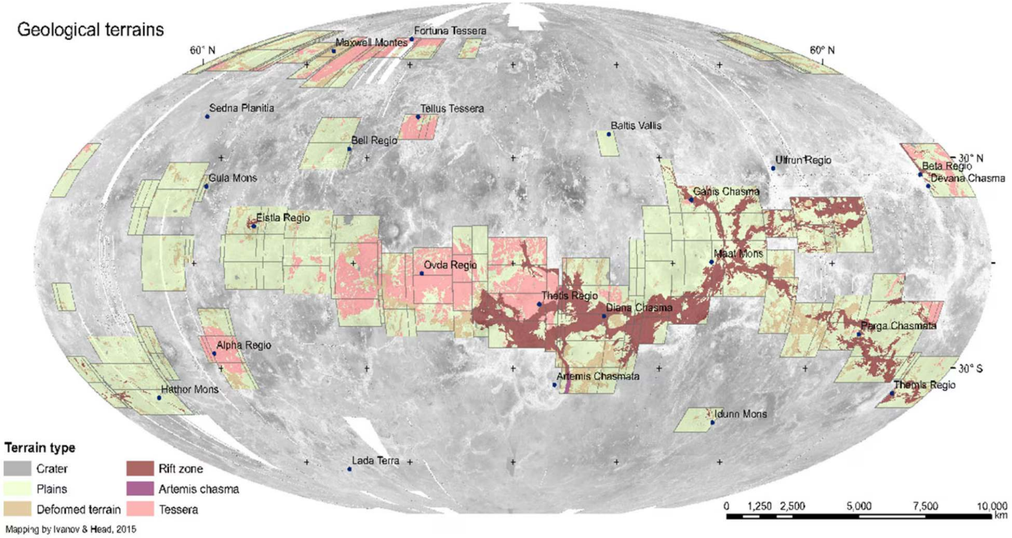


Surface temperature and pressure of Venus, Earth and Mars: the near-twins differ by a factor 92 in pressure and 450 K at the surface. Course-original figure; data from NASA Fact Sheet.



The surface of Venus from Venera 13 (1982): basaltic plates under a 92 bar, 737 K atmosphere. Credit: USSR / NASA NSSDC, public domain.

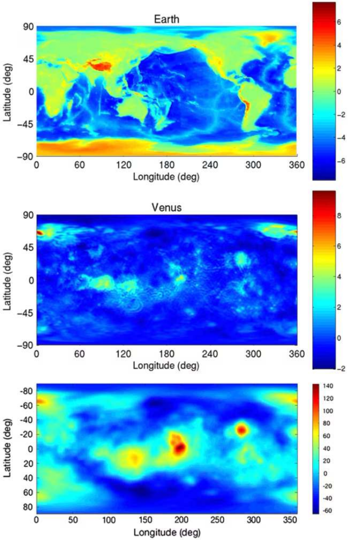
Landers survived two hours, so exploration moved to radar and orbit: Venera (1975–1982) surface images and XRF composition, Magellan (1990–1994) global radar at ~ 100 m, Venus Express and Akatsuki (2006 to present) dynamics, escape rates and super-rotation.



Magellan radar mosaic with terrain overlay: volcanic plains cover ~80% of the surface beside tessera highlands, with no plate boundaries. Widemann et al. (2023), *SSRv* 219, 56.

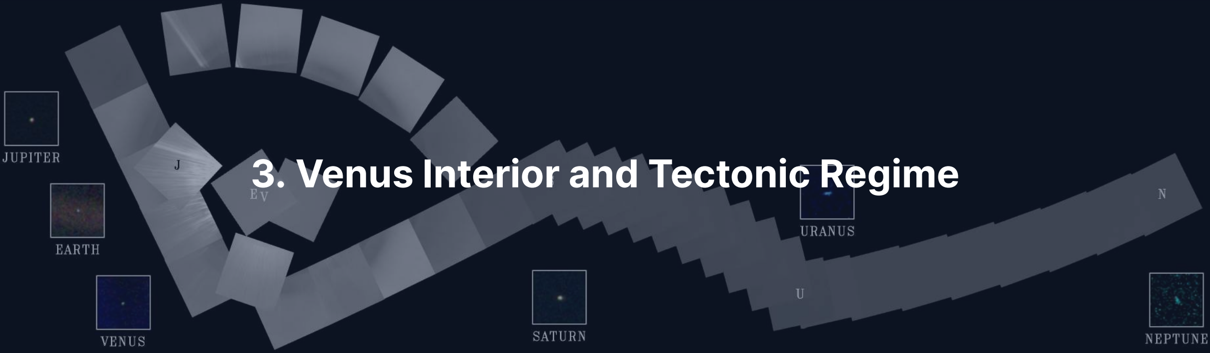
Crater Statistics: A Single Surface Age

- ~ 1000 catalogued impact craters (≥ 2 km) show no spatial clustering
- Uniform density implies a young global surface age of ~ 150–250 Myr
- Debated between catastrophic global resurfacing and steady volcanic equilibrium



Topography and geoid: unimodal hypsometry on Venus against Earth's bimodal ocean-continent distribution rules out Earth-style plate tectonics. Smrekar et al. (2018).

3. Venus Interior and Tectonic Regime

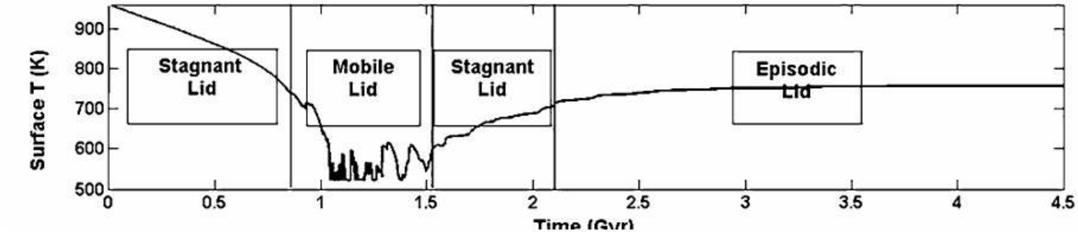


Voyager 1 Solar System family portrait (1990); each of the six planets it detected appears in a small inset box. Credit: NASA/JPL-Caltech

Venus’ Interior: Inferred, Not Measured

Without landed seismology or precession measurements, Venus’ structure is modelled by Earth analogy.

- Moment of inertia factor $C/MR^2 \approx 0.33$ is assumed, not measured
- Inferred metallic core radius $R_c \approx 3000$–3500 km (~ $0.55\,R_V$)
- Mean density of 5243 $\mathrm{kg\,m^{-3}}$ requires a substantial iron core



Surface temperature of Venus in one numerical model through stagnant lid, mobile lid, stagnant lid and episodic lid intervals. Credit: Smrekar et al. (2018).

The alternatives debated for Venus today are a **steady stagnant** lid with localised plumes or **episodic catastrophic overturn**; either way, no Earth-style plate tectonics.

The Missing Magnetic Field

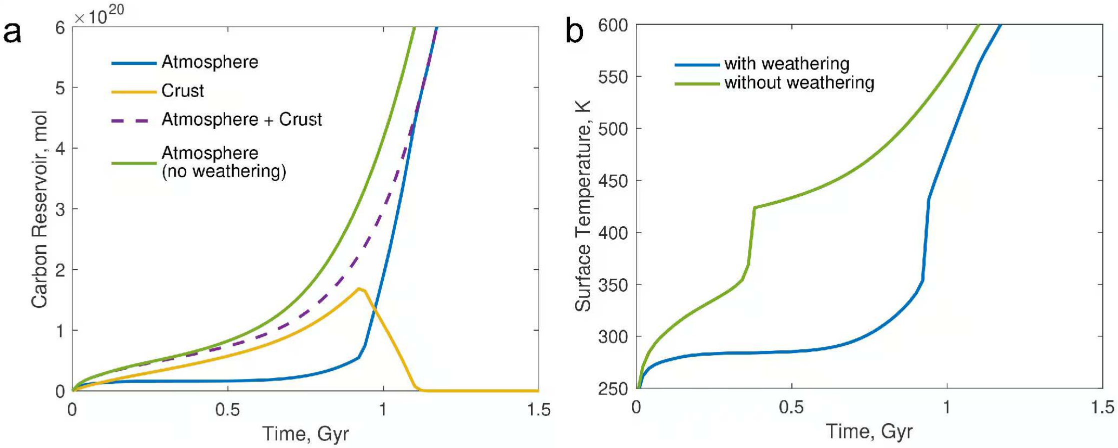
- Internal magnetic field measured at < 10^{-5} times Earth's field
- Stagnant-lid insulation suppresses core heat loss, shutting down thermal convection
- Unshielded atmosphere enables solar-wind induced escape and ion pickup

Why Stagnant Lid?

A hot, dry surface makes a strong lithosphere that resists subduction.

- Plate subduction requires a hydrated, mechanically weakened lithosphere
- High surface temperatures (737 K) promote ductile flow rather than brittle faulting
- No subduction return leg breaks the long-term carbon cycle permanently

Source: Smrekar et al. (2018); Gillmann et al. (2022)

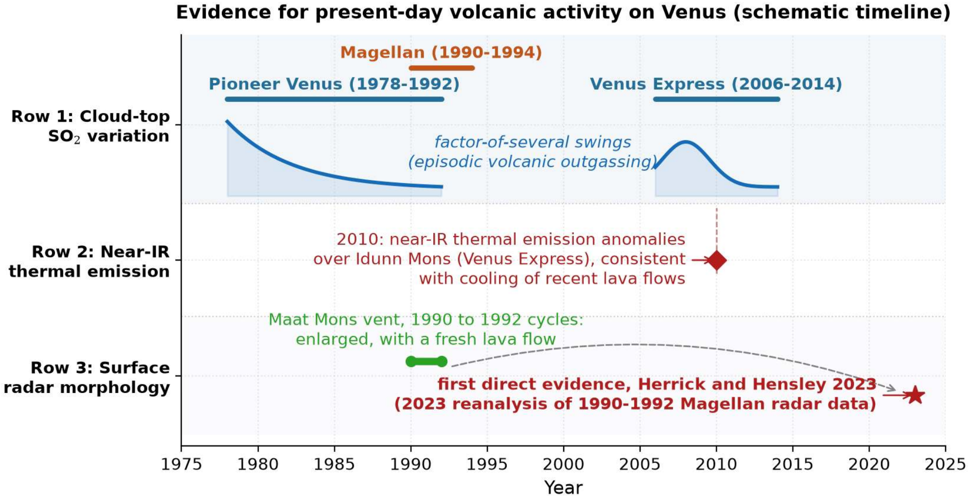


Coupled interior-atmosphere evolution of a stagnant-lid Venus: weathering keeps temperate conditions for 0.9 Gyr until water loss triggers crustal decarbonation to a 92-bar end state.
Credit: Honing et al. (2021).

Is Venus Volcanically Active Today?

- Magellan radar reanalysis reveals a ~ 2 km^2 vent change on Maat Mons (1991–1992)
- Venus Express thermal emissivity anomalies indicate lava flows younger than 250 kyr
- Decadal fluctuations in atmospheric SO_2 suggest episodic volcanic outgassing

Venus remains volcanically active today, losing internal heat through mantle plumes.



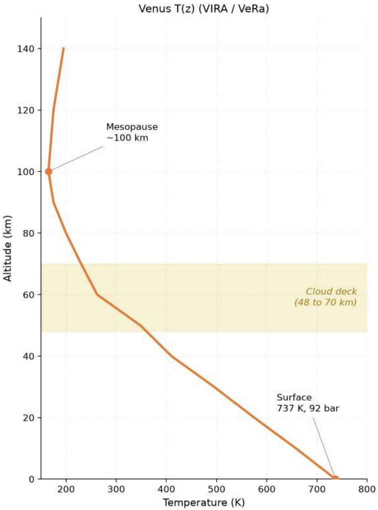
Three lines of evidence for present-day volcanism on one timeline: cloud-top SO_2 swings, Idunn Mons thermal anomalies, and the Maat Mons vent change (Herrick and Hensley 2023). Course figure.

4. Venus Atmosphere and Climate

Earth from Apollo 17 (1972). Credit: NASA / Apollo 17 crew

Venus Atmosphere: Structure

- 96.5% $\mathrm{CO_2}$, 3.5% $\mathrm{N_2}$ at 737 K and 92 bar; $^{40}\mathrm{Ar}/^{36}\mathrm{Ar}$ ≈ 1.1 (Earth ~ 300): Venus has degassed only a fraction of its interior
- Global cloud deck at 48–70 km of ~ 75% sulfuric acid droplets, no tropopause cold trap
- Lower atmosphere on a dry adiabat, Γ ≈ 8 $\mathrm{K\,km^{-1}}$; scale height H ≈ 15.9 km



Venus temperature profile from the 737 K, 92 bar surface through the cloud deck (48–70 km) to 100 km: a monotonic fall with no inversion, close to adiabatic in the deep troposphere. Credit: Pioneer Venus / VIRA and Venus Express; course-original figure.

Atmospheric Super-Rotation

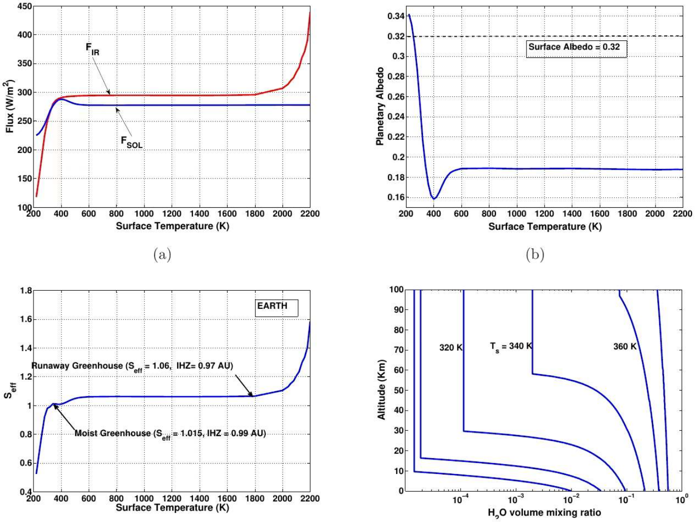
- Cloud-top winds of ~ 100 $\mathrm{m\,s^{-1}}$ circle the planet in ~ 4 days
- Solid-body retrograde rotation takes 243 days (60× angular velocity ratio)
- Maintained by wave transport and atmospheric thermal tides

Venus' Greenhouse: T_{surf} from T_{eq}

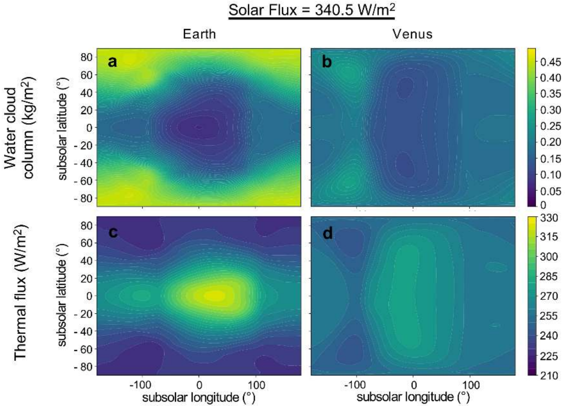
$$T_{\mathrm{eq}} = \left[\frac{S(1-A)}{4\sigma}\right]^{1/4} \approx 227\ \mathrm{K}, \quad T_{\mathrm{surf}} = 737\ \mathrm{K}$$

- Bond albedo $A = 0.77$ reflects most sunlight, yielding $T_{\mathrm{eq}} = 227\ \mathrm{K}$
- One-layer greenhouse $T_s = 2^{1/4}\,T_{\mathrm{eq}} \approx 270\ \mathrm{K}$ falls far short
- Greenhouse warming $\Delta T = +510\ \mathrm{K}$ dwarfs Earth's $+33\ \mathrm{K}$

Driven by 92 bar of $\mathrm{CO_2}$, with $\mathrm{SO_2}$, $\mathrm{H_2O}$ and the clouds closing the spectral windows.

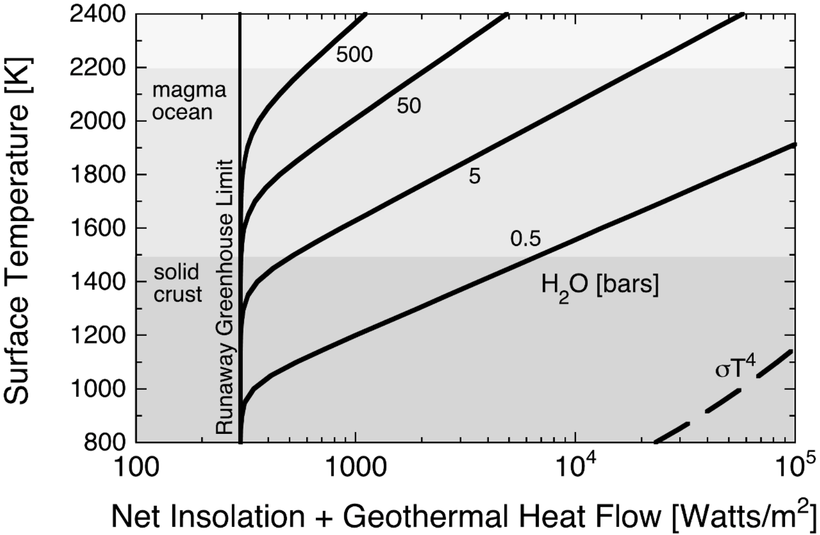


Climate-model runaway threshold: outgoing longwave radiation asymptotes to 282–291 W/m^2 as water vapour closes the infrared windows. Credit: Kopparapu et al. (2013).



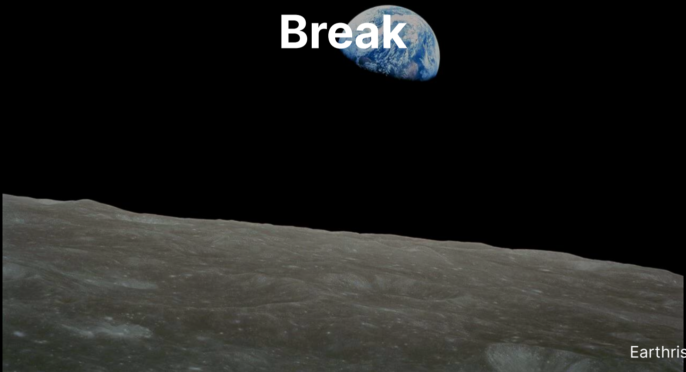
Cloud and thermal emission maps of a hot, steamy Earth (a, c) and Venus (b, d) at the same flux of 340.5 W/m^2. Credit: Turbet et al. (2021), Fig. 2a-d.

Clouds gather on the night side of both although Venus turns 243 times slower, so the pattern does not need slow rotation; night-side clouds trap thermal radiation, and the emission maps anticorrelate with the cloud maps.



Thermal radiation limit and runaway threshold: a saturated atmosphere emits at most ~ 280–300 $\mathrm{W\,m^{-2}}$, beyond which surface water vaporises completely. Zahnle et al. (2007).

Break

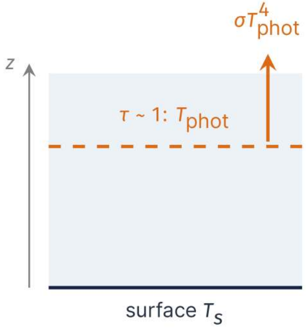


Earthrise, Apollo 8 (1968). Credit: NASA

5. Blackboard Derivation: Simpson-Nakajima Limit

Voyager 1 Solar System family portrait (1990); each of the six planets it detected appears in a small inset box. Credit: NASA/JPL-Caltech

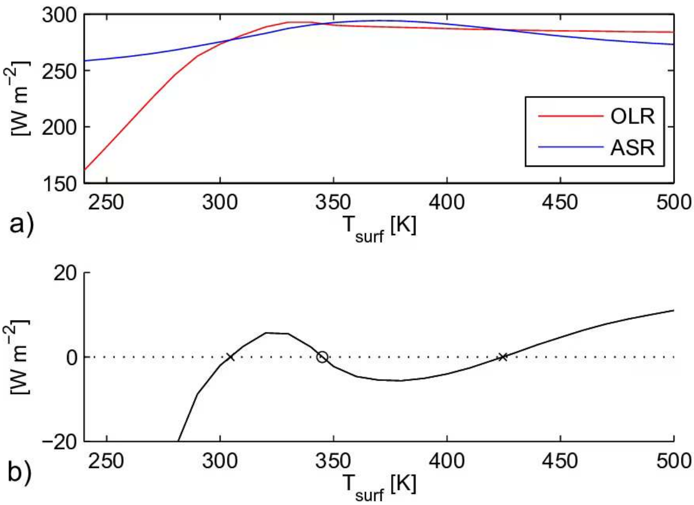
Goal and strategy



Goal: a water-saturated atmosphere has a maximum OLR, independent of surface temperature.

- Saturated everywhere; one grey IR opacity κ; hydrostatic with gravity g
- The photosphere sits where the IR optical depth reaches $\tau \sim 1$
- Clausius-Clapeyron fixes $p_{\text{sat}}(T)$ at every level

Optical depth fixes the photosphere pressure, saturation fixes its temperature.



Outgoing longwave and absorbed shortwave radiation against surface temperature: multiple thermal equilibria in a saturated greenhouse atmosphere. Credit: Wordsworth et al. (2013).

Result: The OLR Asymptote

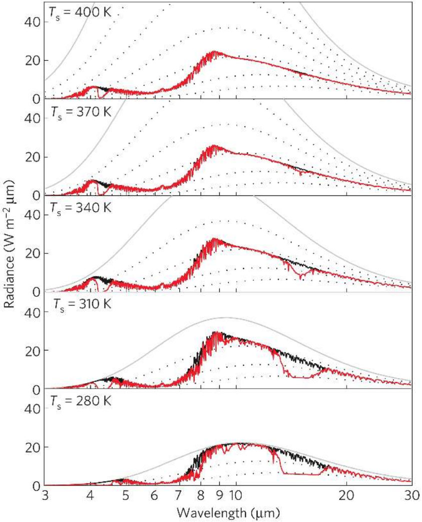
Optical depth unity fixes the photosphere pressure, saturation fixes its temperature (board):

$$p_{\mathrm{phot}} = \frac{g}{\kappa} = p_{\mathrm{sat}}(T_{\mathrm{phot}}) \quad \Rightarrow \quad T_{\mathrm{phot}} \approx 260\ \mathrm{K}$$

$$F_{\mathrm{OLR}}^{\mathrm{max}} \approx \sigma T_{\mathrm{phot}}^4 \approx 5.67 \times 10^{-8} \times (260)^4 \approx 260\,\mathrm{W\,m^{-2}}$$

Refined line-by-line calculations: $F_{\mathrm{OLR}}^{\mathrm{max}} \approx 280–310\ \mathrm{W\,m^{-2}}$ (Kasting 1988; Goldblatt et al. 2013).

As T_s rises, the photosphere migrates upward but its temperature stays clamped: thermal flux saturates.



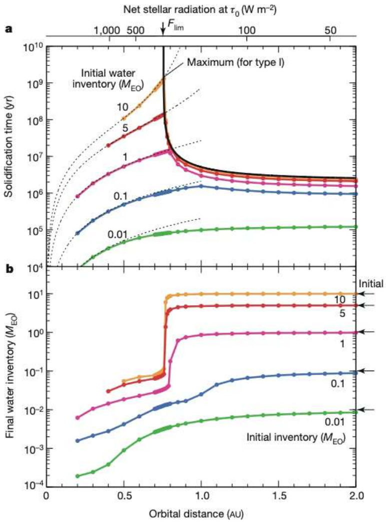
Thermal spectra of a water-rich atmosphere for surface temperatures from 280 to 400 K: the flux saturates at the 282 W/m^2 asymptote as the 8–14 μm window closes. Credit: Goldblatt et al. (2013).

Against the Limit: Earth Below, Venus Above

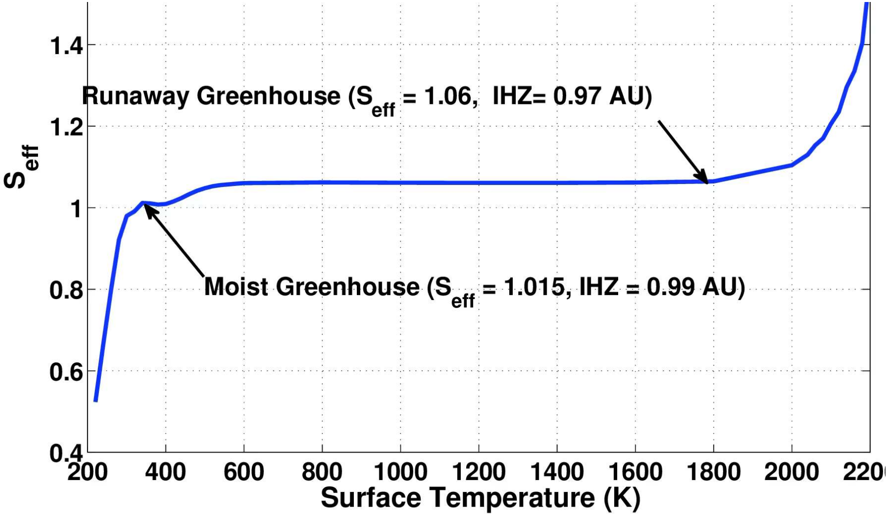
$$F_{\mathrm{abs}} = \frac{S_\odot}{4a^2}(1 - A)$$

- Earth (a = 1 AU, A = 0.30): $F_{\mathrm{abs}} \approx 240\ \mathrm{W\,m^{-2}}$, below $F_{\mathrm{OLR}}^{\mathrm{max}}$
- Early Venus (a = 0.723 AU, Earth-like A): $F_{\mathrm{abs}} \approx 460\ \mathrm{W\,m^{-2}}$, far above it
- Present Venus (A = 0.77) absorbs only ~ 150 $\mathrm{W\,m^{-2}}$: the albedo is the product of the runaway, not its cause

Setting $F_{\mathrm{abs}} = F_{\mathrm{OLR}}^{\mathrm{max}}$ gives the inner edge of the habitable zone: the runaway limit at 0.84–0.97 AU, the moist greenhouse limit at $a_{\mathrm{IHZ}} \approx$ 0.95–0.99 AU.



Solidification time and retained water for Type I and Type II terrestrial planets: beyond the critical distance of 0.77 AU magma oceans solidify quickly and keep their oceans. Credit: Hamano et al. (2013).



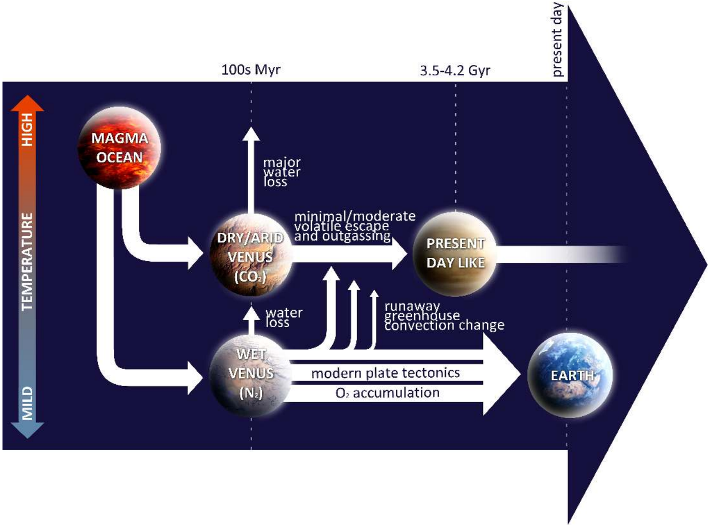
Inner habitable zone: the runaway greenhouse sets the inner edge at S_{eff} = 1.06 (0.97 AU around the present Sun), with Venus (1.91 $S_{\oplus}$) well inside. Kopparapu et al. (2013).

6. Venus Water Loss and the D/H Ratio

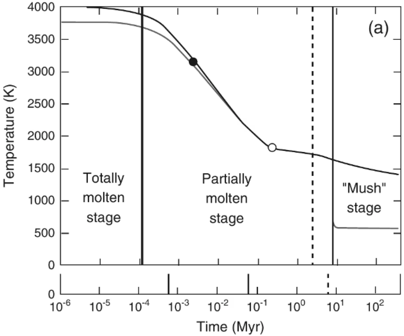
Mercury, Venus, Earth, Mars at approximate relative scale. Credit: NASA/JPL

When Did Venus Lose Its Water?

- **Early loss (Hamano Type II):** the steam atmosphere never condenses; water escapes during the magma-ocean phase
- **Late loss (Way et al. 2016):** a temperate surface persists for Gyr until solar brightening triggers the runaway
- The timing needs noble-gas isotopes and surface mineralogy from the next missions

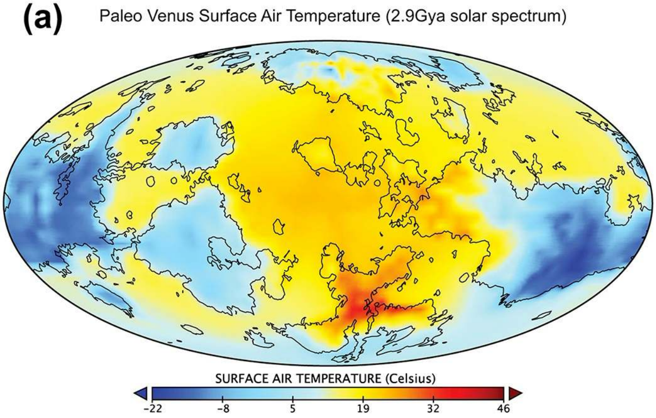


Two end-member branches of Venus' water history: early hydrodynamic loss from a magma ocean (top) against late desiccation after an extended habitable period (bottom). Credit: Gillmann et al. (2022).



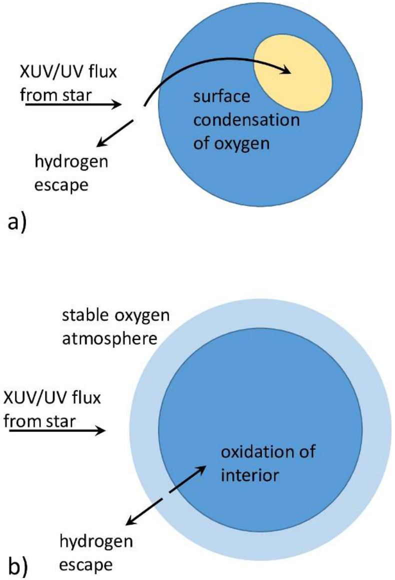
Potential (black) and surface (grey) temperature of a crystallising magma ocean coupled to its steam atmosphere on a Venus-mass planet at Venus's orbit; the dashed line marks 98% solidification. Credit: Lebrun et al. (2013).

Solidification takes about 10 Myr at Venus's orbit (1.5 Myr at Earth's, 0.1 Myr at Mars's), long enough for the steam atmosphere to photolyse and lose hydrogen. Below about 0.66 AU the magma ocean never freezes; Venus at 0.72 AU sits just outside that limit.



Surface temperature of a paleo-Venus at 2.9 Gya in a GCM snapshot, from −22°C at the poles to +46°C at the equator. Credit: Way et al. (2016).

Cloud feedback from slow rotation reflects flux to space; a global mean T < 290 K can persist for several Gyr, with water loss within the last ~ 1 Gyr: a habitable Venus is dynamically possible, not required.



Water loss by photolysis and hydrogen escape, and the fate of the oxygen left behind. Credit: Wordsworth & Pierrehumbert (2014).

UV photolysis ($H_2O + h\nu \rightarrow H + OH$) splits water high up, hydrogen escapes by Jeans and hydrodynamic escape, and the oxygen is absorbed by crustal oxidation or stripped by solar-wind ion pickup.

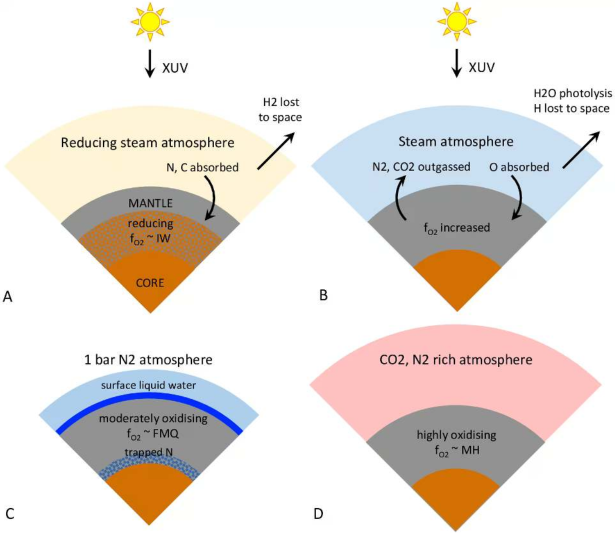
The D/H Ratio: Evidence for Lost Water

Venus' atmospheric D/H is ~ 150× Earth's: the record of massive past hydrogen escape.

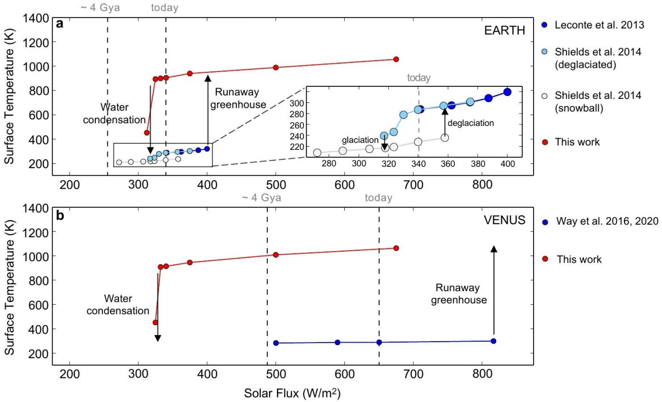
Body	D/H ($\times 10^{-4}$)
Protosolar (H_2)	~ 0.20
Earth (oceans)	1.56
Comets (Halley, 67P)	1.5–5
Venus	~ 240
Mars	~ 8

Source: Donahue et al. (1982); de Bergh et al. (1991)

- Light $^{1}\mathrm{H}$ escapes more easily than $^{2}\mathrm{H}$
- Rayleigh distillation: $R/R_0 = f^{\alpha-1}$
- Mass-ratio limit ($\alpha = 0.5$): $f \approx 150^{-2}$, initial water ~ 450 m GEL; the thermal-velocity limit ($\alpha = 0.71$) gives an unphysical > 100 oceans; early hydrodynamic loss at $\alpha \approx 1$ leaves no D/H trace, so 450 m is a lower bound

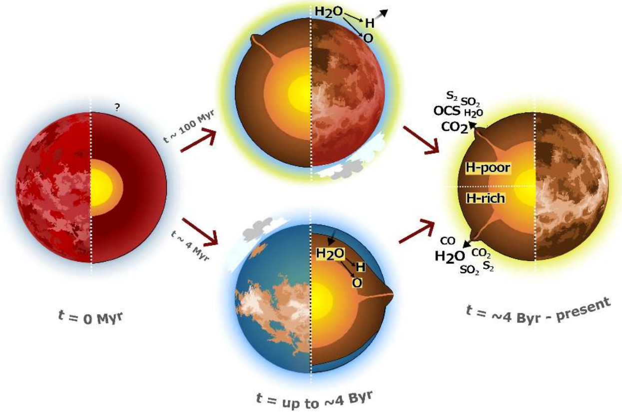


The water-loss redox pump: hydrogen escape oxidises the mantle and leaves a 92-bar $\mathrm{CO_2}$ atmosphere, against Earth's 1-bar $\mathrm{N_2}$ state. Credit: Wordsworth (2016).



Bistability and hysteresis of the surface state against stellar flux. Credit: Turbet et al. (2021), Fig. 4.

For some stellar fluxes a condensed state (ocean, thin atmosphere) and a runaway state (hot, thick steam atmosphere) coexist; the transition is one-way, the steam state has no path back to condensation.

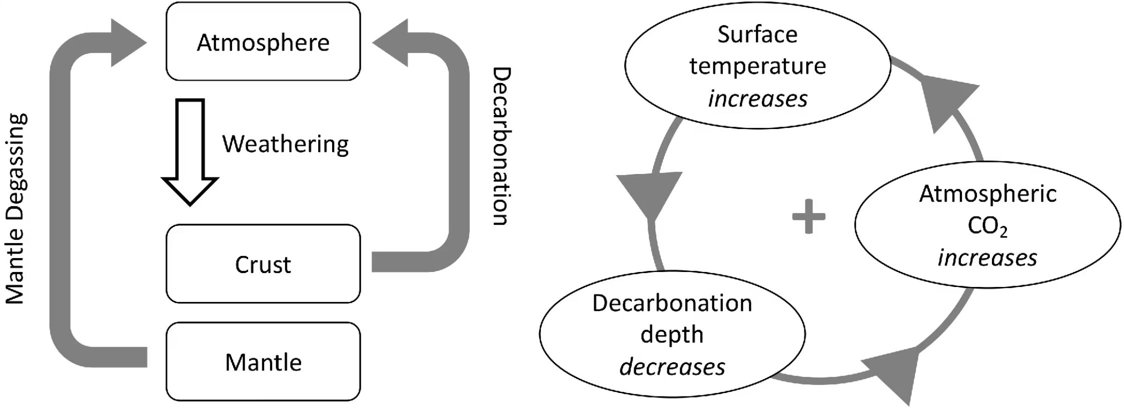


Two end-member pathways for Venus' water history: dry (top) and wet (bottom). Credit: Constantinou et al. (2024).

Dry branch: no surface ocean, early H loss, dry mantle today; wet branch: the ocean condenses, late loss, wet mantle. Current photochemistry favours the dry branch (volcanic $H_2O \lesssim 6\%$); DAVINCI noble gases will discriminate.

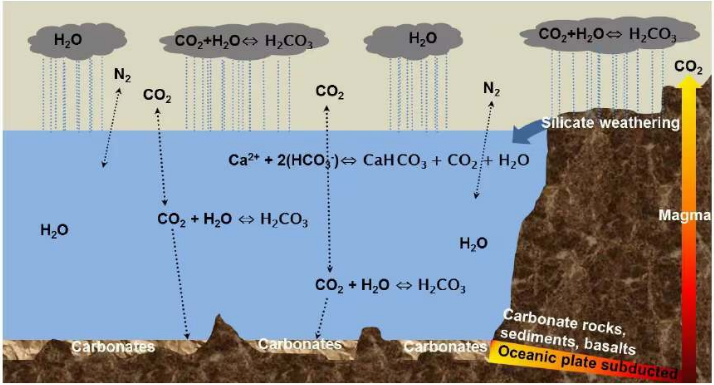
7. Why Did Earth and Venus Diverge?

Mercury, Venus, Earth, Mars at approximate relative scale. Credit: NASA/JPL



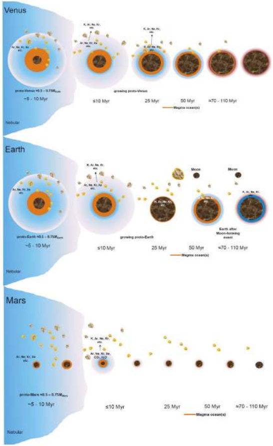
The carbon cycle of a stagnant-lid Venus and where it breaks. Credit: Honing et al. (2021).

Without liquid surface water the silicate weathering sink collapses, and without subduction no carbonate returns to the mantle: volcanic outgassing drives $\mathrm{CO_2}$ to the observed 92 bar.



Earth's closed carbon loop: volcanic outgassing, silicate weathering, carbonate sedimentation and subduction. Credit: Lammer et al. (2018).

Weathering is the temperature-dependent sink, subduction the return leg; Venus today has neither a liquid-water sink nor a subduction return, so both halves of the loop are broken.



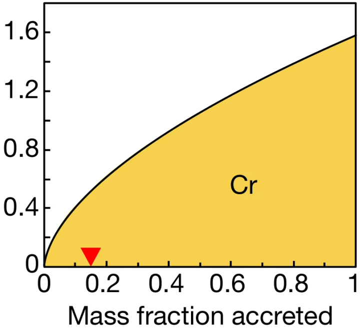
Comparative accretion of Venus, Earth and Mars. Credit: Lammer et al. (2018).

The initial water and carbon inventories overlap substantially: the divergence is not from initial conditions alone, early evolution decides which branch each planet takes.

Why the Twins Diverged: Habitability Implications

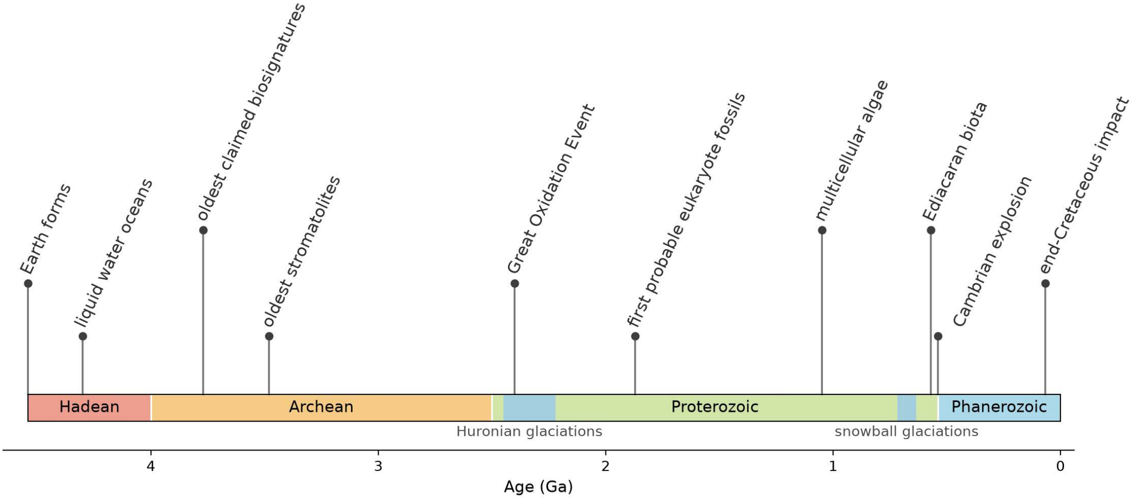
1. **Solar flux:** 1.91× Earth's insolation, above the runaway threshold
2. **Geophysics:** no intrinsic dynamo, slow retrograde rotation
3. **Tectonics:** a stagnant lid removes the subduction return leg of the carbon cycle

Venus defines the inner edge of the habitable zone: coupled interior-atmosphere feedbacks decide whether a terrestrial planet keeps its water.



Probability density for the chromium-bearing fraction of Earth's accreting mass against the accreted mass fraction: Cr arrived nearly mass-proportionally, unlike the back-loaded Mo and Ru. Credit: Dauphas (2017).

With the O, Ti, Ni, Mo and Ru tracers this gives three stages: the first 60% of Earth's mass about 51% enstatite-like, 40% ordinary-chondrite and 9% carbonaceous, the rest enstatite-like. The carbonaceous component, and most of Earth's water, came early rather than in the late veneer.



Milestones in the history of life across Earth's 4.54 Gyr, a path of continuous habitability denied to Venus. Credit: Course-original figure.

Modern stromatolites at Shark Bay, Western Australia; fossil stromatolites of the same construction in the Pilbara, dated to 3.48 Ga, are the oldest widely accepted evidence for life. Photo by Paul Harrison (Wikimedia Commons), CC BY-SA 3.0.

8. Recent Advances and Future Missions

VERITAS, DAVINCI and EnVision over an Akatsuki UV view of Venus. Credit: Widemann et al. (2023)

The 2030s Venus fleet, DAVINCI and VERITAS (NASA) and EnVision (ESA): DAVINCI's descent probe measures the noble gases that separate early from late water loss. Widemann et al. (2023).

Summary

1. **Bulk twin:** same size and density, but ~ 500 K and 92 bar apart; the **runaway limit** of Simpson-Nakajima (280–310 $\mathrm{W\,m^{-2}}$) sets the inner habitable zone
2. **D/H evidence:** ~ 150× Earth's ratio records massive water loss
3. **Broken cycle:** no surface water stops the weathering sink; a stagnant lid removes the subduction return leg

Next: Lecture 10 Mercury and Mars.

Questions?

Next: **Lecture 10. Rocky Planets II: Mercury & Mars**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`